

STARFALL WORKS

Wargame Counter Creator

User Manual - Version 0.29

Design individual counters. Build two-sided sets. Arrange printable sheets. Export production-ready PDFs.

Local browser application - no installation or server required

Prepared for Wargame Counter Creator v0.29

How to Use This Manual

This manual is organized around the normal production workflow: create counters, refine their appearance, define front/back behavior, arrange them on sheets and export files for printing. You can read it front to back or jump directly to the section you need.

Section	Use it when you want to...
1. Getting Started	Open the application and understand the workspace.
2. Counter Designer	Create and edit individual counter designs.
3. Templates and Artwork	Choose Classic, Six-Value or Information layouts and add symbols.
4. Styling and Typography	Control colors, stripes, text sizes, numbers and highlights.
5. Two-Sided Counters	Design paired fronts and backs that stay registered.
6. Managing a Counter Set	Use quantity, duplicate, reorder, save and export CSV.
7. Sheet Designer	Set paper, orientation, margins, gutters and guides.
8. General Double-Sided PDF	Create a direct duplex-ready Letter/A4 PDF.
9. Superior POD Export	Create the manufacturer-specific 18 x 12 inch POD file.
10. Manual POD Planner	Place specific counters in specific POD slots and optimize colors.
11. Printing	Print at true physical size and avoid scaling problems.
12. Troubleshooting	Resolve common layout, sizing and export issues.

Version note. This manual describes Wargame Counter Creator v0.29. Labels and behavior may change in later versions.

1. Getting Started

1.1 Launching the application

1. Extract the Wargame Counter Creator ZIP file to a folder on your computer.
2. Open the extracted folder.
3. Double-click index.html, or open it from Chrome, Edge, Firefox or Safari.
4. The application runs locally in the browser. No server or installation is required.

Recommended browser. A current desktop version of Chrome or Edge provides the most predictable experience for large PDF exports and manual POD layouts.

1.2 Main workspace

The application has two primary workspaces:

- Counter Designer - creates and edits individual counter designs.
- Sheet Designer - arranges counters for print, general PDF export and Superior POD export.

The main toolbar includes New, Save Project, Export CSV, project loading, General Double-Sided PDF, Superior POD PDF and browser Print/PDF functions.

1.3 The Counter Designer layout

Area	Purpose
Left: Counters	Lists every counter design in production order. Add, duplicate, delete and reorder counters here.
Center: Preview	Shows the selected counter at an enlarged zoom. Optional guides show bleed and safe-area boundaries.
Right: Properties	Controls template, size, quantity, colors, fonts, artwork, values, bleed, safe inset and two-sided settings.

2. Counter Designer

2.1 Creating a new counter

1. Click the + button in the Counters panel.
2. Select the new counter in the list.
3. Choose a Template and Counter size.
4. Enter the unit name, type and values that apply to the template.
5. Choose colors, silhouette or imported artwork.
6. Set Quantity to the number of physical copies you want printed.

Live editing. Most property changes update the preview immediately. Use the Zoom control when you need to inspect small type, highlights or symbol placement.

2.2 Counter sizes

Preset	Finished size
1/2 inch	0.500 x 0.500 in
5/8 inch	0.625 x 0.625 in
3/4 inch	0.750 x 0.750 in
1 inch	1.000 x 1.000 in

Counter size is a physical measurement used by the direct PDF exporters. For example, a 5/8 inch counter occupies 45 PDF points because PDF coordinates use 72 points per inch.

2.3 Bleed and safe area

Bleed extends background artwork beyond the finished counter edge so minor cutting variation does not reveal white paper. The safe inset identifies an interior area where important text and numbers are less likely to be affected by cutting tolerance.

- Bleed presets: None, 1/32 inch, 1/16 inch and 1/8 inch.
- Safe inset presets: 0.025 inch, 0.040 inch and 1/16 inch.
- Use Show guides in the Counter Designer to see the current guide boundaries.

3. Counter Templates and Artwork

3.1 Classic template

Classic is the traditional combat-unit layout. It provides a unit name, unit type, centered symbol and three bottom values.

Position	Typical use
Bottom left	Attack or first combat value
Bottom center	Defense or second combat value
Bottom right	Movement or third combat value

3.2 Six-Value template

Six-Value adds optional top-left and top-right values while retaining three bottom values. The silhouette occupies its own image zone and the piece name is placed in a separate band below it.

- Top-left and top-right values may be left blank.
- Bottom-left, bottom-center and bottom-right values may also be left blank.
- Each numeric position has independent text color and optional highlight settings.

3.3 Information template

Information counters are intended for markers, reminders, phases, weather, control markers and other non-unit pieces. They emphasize centered text and may also use a built-in or imported image.

3.4 Built-in silhouettes

The built-in silhouette list includes None, Infantry, Tank, Artillery, Truck, Fighter, Bomber, Destroyer, Cruiser, Battleship, Carrier, Submarine and Helicopter.

3.5 Importing custom symbols

1. Use the Imported symbol control in the Properties panel.
2. Import a PNG, JPG or SVG file.
3. Select the imported symbol for the current counter.
4. Use Remove Imported Symbol if it is no longer needed.

Imported artwork is stored as data inside the saved project JSON, which makes the project portable without keeping separate image files beside it.

4. Styling, Fonts and Numbers

4.1 Colors

- Background - main counter field color.
- Border - finished counter edge color.
- Text - default label color.
- Per-number text colors - separate color for each numeric position.
- Per-number highlight colors - separate highlight color for each numeric position.

4.2 Background stripes

Every counter template supports an optional stripe behind the artwork and text.

Stripe	Position options
Vertical	Left, Center or Right
Horizontal	Top, Center or Bottom
None	No stripe

Stripe color is independent of the main background color. This is useful for nationality, formation, status or game-system coding.

4.3 Font size controls

Control	Affects	Range
Label font size (%)	Unit name, unit type and Information text	50% to 200%
Number font size (%)	Top and bottom numeric values	50% to 200%

PDF fitting behavior. The General Double-Sided PDF preserves the requested size whenever it fits. If a label or number would exceed its assigned area, the PDF renderer reduces it only as much as necessary and clips all artwork to the finished counter square.

4.4 Number highlights

Each numeric position can be emphasized independently. Turn on Highlight for a position and choose a highlight color. Highlights remain centered with the number even when font size changes.

5. Two-Sided Counters

Two-sidedness is defined on the counter itself. This keeps the front and back paired throughout quantity expansion, sheet layout and PDF export.

5.1 Creating a two-sided counter

1. Select the counter.
2. Turn on Two-sided counter.
3. Use Front and Back to switch between the two faces.
4. Optional: click Copy Front to Back to use the front as the starting point.
5. Edit the back independently.

The front and back share physical size, bleed and safe-area geometry so they remain registered. Other visual/content properties may differ between the faces, including template, colors, labels, values, symbol, stripe, highlights and font sizes.

5.2 One-sided counters

A one-sided counter does not require a separate back design. In direct PDF exports, its back uses the front background treatment without the front text, symbol or numeric values. This provides a clean reverse side while maintaining visual continuity.

6. Managing a Counter Set

6.1 Quantity

Quantity defines how many physical copies of a design are produced. A single counter record with Quantity 8 becomes eight copies in sheet and PDF output. For a two-sided counter, all eight copies receive the correct paired back.

6.2 Duplicate and delete

Duplicate is useful when building families of similar counters. Delete removes the selected design from the project.

6.3 Reordering

Use the up and down arrows beside counters in the left-hand list. The list order is the production order used by automatic sheet and PDF layouts. Quantity expansion happens after ordering, so copies of a design remain grouped together.

6.4 Save and load projects

Save Project writes the current work to a JSON project file. The project preserves counter definitions, imported symbols, two-sided data and Superior POD manual placement information. Load the project file later to continue working.

6.5 Export CSV

Export CSV writes one row per counter design rather than one row per physical copy. The export includes the printed values and the styling information needed to document the design.

- Template, size and quantity
- Unit name, type and Information text
- Background, stripe, border and text colors
- Label and number font-size percentages
- Numeric values, number colors and highlights
- Symbol/image reference
- Two-sided data, including back-side fields where applicable

7. Sheet Designer

The Sheet Designer controls general paper layout and the Superior POD workflow.

7.1 General sheet setup

Setting	Options / purpose
Paper	US Letter or A4
Orientation	Portrait or Landscape
Margin	0.25, 0.375 or 0.50 inch

Setting	Options / purpose
Gutter	None, 1/32, 1/16 or 1/8 inch
Crop marks	Shows cutting marks in the sheet layout
Bleed guides	Shows bleed boundaries in the sheet preview

Use Refresh Layout after changing sheet settings if the preview does not immediately reflect the new arrangement.

8. General Double-Sided PDF

General Double-Sided PDF is the preferred direct export when you are not using the Superior POD manufacturer format. It creates the PDF directly rather than opening the browser print dialog.

8.1 What the exporter does

- Uses the selected Letter/A4 paper size and orientation.
- Honors margin and gutter settings.
- Uses the current counter-list order and expands quantities.
- Supports mixed counter sizes.
- Creates each front page followed immediately by the matching back page.
- Mirrors the back page horizontally for duplex registration.
- Uses true designed backs for two-sided counters.
- Uses background-only treatment for one-sided backs.
- Writes PDF pages at true physical dimensions using 72 PDF points per inch.

Important printing setting. Print the generated PDF at Actual Size / 100%. Do not use Fit, Shrink oversized pages or Scale to page if you need the specified counter dimensions to remain exact.

8.2 True-size example

A 5/8 inch counter is 0.625 inches wide. In the finished PDF that equals 45 PDF points (0.625 x 72). The renderer uses a 300 DPI raster internally for image quality but the page itself retains its true physical size.

9. Superior POD PDF Export

The Superior POD exporter is a manufacturer-specific format for 5/8 inch counters. It uses the supplied 18 x 12 inch press-sheet geometry and 176 front positions per sheet.

9.1 Requirements and behavior

- Counters assigned to Superior POD output must use the 5/8 inch counter size.
- The PDF page is 18 x 12 inches at true physical size.
- Front positions follow the manufacturer slot geometry.
- Back positions are horizontally mirrored to the manufacturer layout.

- Automatic backs use designed back artwork for two-sided counters and front background treatment for one-sided counters.
- Repeat fronts (mirrored) is available as an alternate back mode.
- The final PDF does not include the colored template guide layers.

9.2 Bleed behavior

The POD exporter uses color-aware bleed between occupied counter positions:

- If all counters use the same background color, that color can flow across the entire press sheet.
- If neighboring counters have the same background color, the shared gap becomes one continuous color field.
- If neighboring counters use different colors, each color extends halfway into the gap and the two colors meet at the midpoint.
- The same bleed logic is applied to linked back positions.

10. Manual Superior POD Layout Planner

Manual slot layout lets you decide exactly which counter occupies each manufacturer front position. The on-screen planner uses the same coordinates as the exported PDF, so the arrangement you see is the arrangement the exporter uses.

10.1 Choosing a POD layout mode

Mode	Behavior
Automatic counter order	Fills POD positions from the left-hand counter-list order.
Manual slot layout	Lets you place specific counter designs into specific front positions.

10.2 Manual planner controls

- Fill from Counter Order - seeds the manual sheet from the current list order.
- Group by Color - creates a starting arrangement grouped by background/stripe treatment.
- Add POD Sheet - adds another 176-slot press sheet.
- Clear This Sheet - removes assignments from the active sheet.
- Remove This Sheet - removes the active press sheet.
- Selected slot - click a front slot, then use the dropdown to assign or change its counter.
- Clear Slot - empties the selected front position.

The quantity summary shows assigned versus required copies for every counter. Missing or over-assigned counters are flagged. Blank slots are allowed.

Back-side safety. You only place front counters manually. The corresponding back remains linked and is placed automatically in the correct mirrored back position, so manual layout cannot mismatch a front and back.

10.3 Color optimization workflow

1. Switch Superior POD layout to Manual slot layout.
2. Click Group by Color for a useful starting arrangement.
3. Inspect rows and columns in the exact press-sheet workspace.
4. Select individual slots and move designs as needed to create larger same-color neighborhoods.
5. Watch the assigned/required quantity summary while making changes.
6. Export Superior POD PDF and review the resulting sheet before submitting it to the printer.

11. Printing and Production Guidance

11.1 General PDF printing checklist

- Use Actual Size or 100% scale.
- Disable Fit to page / Shrink oversized pages / Scale to printable area.
- Use duplex printing only if your printer can register front and back accurately.
- For the application-generated back pages, use the duplex flip setting appropriate to your printer and test one sheet before a full run.
- Verify one known counter with a ruler before printing a large batch.

11.2 Browser Print / PDF

Browser Print / PDF is retained for convenience, but browser print dialogs vary and may scale content unexpectedly. For repeatable production output, use General Double-Sided PDF or Superior POD PDF whenever possible.

11.3 Cutting considerations

- Keep essential text and numbers inside the safe area.
- Use bleed where backgrounds reach the counter edge.
- Avoid very thin borders if cutting tolerance is uncertain.
- Use crop marks on general sheets when cutting by hand or with a straightedge.

12. Troubleshooting

Symptom	What to check
Counter prints too large or too small	Print the direct PDF at Actual Size / 100%. Confirm Fit/Scale options are disabled.
Text appears outside a counter in General PDF	v0.29 clips all artwork to the finished square and auto-fits overflowing text. Confirm you are using v0.29 or later.
Numbers look off-center on screen	Use v0.24 or later, which centers values in fixed-width positioning boxes.

Symptom	What to check
POD manual assignment seems wrong	Use the exact POD workspace introduced in v0.20. Click the front slot, then assign through Selected slot.
POD export rejects a counter	Superior POD requires 5/8 inch counters.
Manual POD export reports missing copies	Assign the remaining required copies or confirm that you want to export only the currently assigned counters.
Manual POD export reports too many copies	Remove duplicate slot assignments until assigned quantity does not exceed the counter quantity.
Imported image missing after reopening project	Reload the saved JSON project. Imported symbols are stored in the project file.
Browser print output differs from preview	Use General Double-Sided PDF instead of browser printing for deterministic output.

13. Recommended Production Workflow

1. Create one design for each distinct counter type.
2. Choose the physical size before fine-tuning typography.
3. Set colors, stripe and artwork.
4. Set label and number font sizes while viewing the counter at high zoom.
5. Set quantity instead of duplicating identical counters.
6. For reversible pieces, enable Two-sided and complete the back face.
7. Use the counter-list arrows to establish production order.
8. Save the project before sheet work.
9. For home/general printing, configure Sheet Designer and use General Double-Sided PDF.
10. For Superior POD, switch to the required 5/8 inch size and choose automatic or manual POD layout.
11. If optimizing ink/bleed neighborhoods, use Group by Color and then manually refine slots.
12. Export the PDF, inspect it at 100% and verify one physical dimension before production.
13. Save the final project JSON beside the production PDF for future revisions.

14. Quick Reference

Task	Where to do it
Add a counter	Counter Designer > Counters > +
Duplicate a counter	Counter Designer > Duplicate

Task	Where to do it
Reorder counters	Up/down arrows in the left counter list
Set print quantity	Properties > Quantity
Make a counter reversible	Properties > Two-sided counter
Edit the reverse	Properties > Back
Copy front artwork to reverse	Copy Front to Back
Add custom art	Properties > Imported symbol
Add a colored stripe	Background stripe + Stripe position + Stripe color
Resize labels	Label font size (%)
Resize numbers	Number font size (%)
Save work	Save Project
Document counter data	Export CSV
Create ordinary duplex PDF	General Double-Sided PDF
Create printer-specific sheet	Superior POD PDF
Manually place POD counters	Sheet Designer > Superior POD layout > Manual slot layout
Print via browser	Browser Print / PDF

Best practice. Treat the saved project JSON as the editable source file and the generated PDF as the production output. Keep both together when archiving a game project.

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A Starfall Works product